Mohsin Ali Mirza

Bremen, Germany
+4915207955670 | mohsinalimirxa@gmail.com | linkedin.com/in/mohsin-ali-mirza | github.com/mohsin-ali-mirza

Profile

AI Engineer with 1+ years of experience architecting and deploying end-to-end intelligent systems across the machine learning lifecycle. Specializes in multimodal Generative AI, agentic workflows, and computer vision pipelines, with technical expertise in PyTorch, LangChain, and orchestrating scalable microservices via Docker and Kubernetes. Leverages a unique spatial computing edge in Unity3D to bridge the gap between digital simulation and real-world robotics deployment.

Education

University Of Bremen — Bremen, Germany
Masters of Science in Artificial Intelligence and Intelligent Systems — *1.2 German Grade*

National University of Computer and Emerging Sciences — Karachi, Pakistan
Bachelor of Science in Computer Science CGPA:3.81 / 4.00 — *1.2 German Grade*

Experience

Extended Reality (XR) and AI Developer — Dec 2024 – Present
BearingPoint GmbH — *Hamburg, Germany*

- Architected an AI-driven Virtual Assistant in Unity with RAG pipeline to surface safety compliance insights in <500ms latency, enabling field teams to access real-time regulatory guidance without context-switching.
- Implemented state-of-the-art Speech-to-Text (STT) and Text-to-Speech (TTS) models, optimizing inference latency to ensure seamless, low-latency audio interactions in an immersive XR environment.

Game Developer — Aug 2024 – Nov 2024
Pixel Dream Studios — *Karachi, Pakistan*

- Implemented the Strategy Pattern in Unity to decouple and extend the game's core ability system, enabling seamless integration of diverse player powers like dashing, warping, and blasting on a shipped Steam title.
- Engineered state-driven AI behaviors for 1v1 and 1v4 single-player challenges, ensuring fast paced matches.
- **Link: store.steampowered.com/app/3630900/Snatch/**

AI Developer — Jun 2023 – Aug 2023
Syslab.Ai — *Karachi, Pakistan*

- Developed a scalable, production-ready Movie Recommendation System using a hybrid recommendation model, boosting Precision@10 and Recall@10 by 80% through hyperparameter tuning.
- Containerized and orchestrated the microservices architecture using Docker and Kubernetes, ensuring high availability and seamless horizontal scaling for real-time inference workloads.

Projects

Real-Time 6D Pose Estimation for Car Disassembly | *Computer Vision, Yolov8, ROS2*

- Designed a complete vision-to-action pipeline using ROS 2 Jazzy and Zenoh, integrating YOLOv8 object detection with 6D pose estimation for automated robotic tasks, achieving 85% on real-world parts.
- Implemented an Action Server architecture to manage complex task sequences (e.g., screw detection, subdoor pose estimation) across Kinova Gen3 and RealSense hardware.
- **Link: github.com/Robots4Sustainability/perception**

Face Recognition Using Generative AI | *Computer Vision, Yolov8, Tensorflow*

- Utilized YOLOv8 and face recognition models to detect and capture single or multiple faces from various camera sources, supporting both real-time and batch processing
- Engineered and trained multiple GAN models to generate realistic facial variations, enhancing the accuracy of individual attendance tracking
- **Link: github.com/Mohsin-Ali-Mirza/Multiple-Face-Recognition**

Sign Language Detection | *Computer Vision, LSTM, Tensorflow*

- Engineered and trained an LSTM-based Sign Language Detection using TensorFlow facilitating communication for the deaf and hard of hearing.
- Preprocessed sign language video data by extracting frames and applying sequence labeling for improved LSTM model training, enhancing performance
- **Link: github.com/Mohsin-Ali-Mirza/Sign-Language-Detection-LSTM**

Abstractive Text Summarization | *Transformers*

- Fine-tuned Transformer models (T5, BART, Pegasus) via the Hugging Face Trainer API for abstractive text summarization, implementing custom preprocessing pipelines and ROUGE metric evaluation
- Built an interactive Streamlit dashboard enabling multi-model checkpoint toggling for real-time comparative inference analysis and latency benchmarking.
- **Link: github.com/Mohsin-Ali-Mirza/abstractive-summarization**

Cycle GANS Style Transfer | *Tensorflow, Deep Learning*

- Achieved stable adversarial convergence over 30 epochs, maintaining discriminator losses near the optimal 0.50 baseline (0.49–0.54) while steadily reducing peak generator losses to a balanced 2.8.
- Evaluated and tuned training dynamics to prevent mode collapse, successfully maintaining a healthy minimax equilibrium where both generator and discriminator networks converged symmetrically without saturation.
- **Link: github.com/Mohsin-Ali-Mirza/Cycle-Gans-Style-Transfer/tree/main**

Technical Skills

Technical Skills: C, C++, C #, Python, MLOPS, SQL (Postgres), MonogoDB(NoSQL), HTML/CSS
Frameworks: PyTorch, Tensorflow, LangChain, FastAPI
GameEngines and Tools: Unity3D, Unreal Engine 5, Blender
Robotics: ROS2, ISAAC Sim
Augmented and Virtual Reality Frameworks: AR Foundation, Vuforia, Oculus, OpenXR
Developer Tools: Git, Gitlab, Docker, Kafka, Grafana, Nginx, AWS, Azure, RESTful APIs, Microservices, CI/CD Pipelines

Awards

Participation Agentic AI Hackathon — BMW	Mar 2026
Winner AI Toolbox Hackathon — U Bremen Alliance	Dec 2025
Participation KI Hackhathon — Digital University Hub	Sep 2024
Gold Medalist x2 — FAST Bachelors	Jun 2024
Winner Data Science — DevDay FAST	Apr 2024
Winner Game Jam — Procom	Apr 2022

Certifications

AWS Cloud Technical Essentials — Amazon	Oct 2023
Introduction to TensorFlow for AI, ML, and Deep Learning — DeepLearning.Ai	Oct 2023
Introduction To Machine Learning In Production — Deeplearning.Ai	Sep 2023
Generative Ai — Google	Aug 2023